

External transport of donor-level SLE classifiers across two cohorts: corrected Geneformer V1 and pseudobulk analysis

Abstract

Objective. We evaluated donor-level SLE versus healthy-control discrimination after development in an adult cohort and application to an external cohort with mostly source-classified Children samples. This report is a versioned correction of the Geneformer V1 analysis.

Methods. We analyzed 261 development donors (162 SLE, 99 healthy; 1,263,676 cells) and 56 external samples (40 SLE, 16 healthy; 363,083 cells). Corrected V1 cell embeddings used explicit V1 tokenization, shared-gene input, penultimate-layer cell extraction, and donor mean pooling. We repeated nested donor cross-validation and full-development fitting. External AUROCs were compared by paired DeLong tests with Holm correction and paired donor-bootstrap intervals.

Results. Corrected shared-gene Geneformer AUROC was 0.9746 in development nested cross-validation and 0.7344 (95% donor-bootstrap CI 0.6017–0.8597) externally. External pseudobulk and age AUROCs were 0.8984 and 0.5781. Geneformer minus age was +0.1563 (95% CI –0.0859 to +0.4040; Holm $p = 0.2046$); Geneformer minus pseudobulk was –0.1641 (95% CI –0.2916 to –0.0384; Holm $p = 0.0175$). Neither prespecified Geneformer-superiority criterion was met. Native-input sensitivity yielded Geneformer AUROC 0.7484.

Conclusion. In this previously examined, distribution-shifted external cohort, corrected V1 features did not meet either prespecified Geneformer-superiority criterion. The findings are specific to this task, model version, feature construction, and two cohorts; they do not establish clinical diagnostic performance or general model inferiority.

Keywords: single-cell RNA sequencing; foundation models; Geneformer; external validation; preregistration; systemic lupus erythematosus; benchmarking

1. Introduction

Large pretrained models have been adapted to single-cell transcriptomics. Geneformer was pretrained on roughly 30 million single-cell transcriptomes [1], scGPT on more than 33 million cells [2], and other examples include scFoundation [3] and scBERT [4]. Downstream performance is often evaluated within the same cohort by cross-validation, which estimates performance in the development population but does not directly test transfer to an independently collected cohort.

Training and test folds from one cohort share technical and population characteristics such as sequencing chemistry, sample handling, recruitment setting, and demographic composition. Cohort-associated structure can therefore persist across folds even when it does not transfer to another dataset. The distinction between internal and independent external validation is well established in clinical prediction research [5,6].

We studied donor-level SLE discrimination in the adult Perez et al. development cohort [7] and GSE135779 from Nehar-Belaid et al. [8]. The task, three arms, two co-primary comparisons, Holm procedure, and decision rule were preregistered before development data were loaded. The historical external evaluation was later found to use an unsupported Geneformer V1 extraction configuration and an invalid equal-AUROC test. This paper reports a versioned correction with the originally specified shared-gene input as the main analysis, model-native input as sensitivity, and explicit paired inference. Because the external cohort had already been examined, the correction is a reanalysis rather than a new untouched confirmatory validation.

Neither co-primary superiority comparison met the prespecified criterion. Lower external AUROC point estimates were observed for Geneformer, pseudobulk, and age, so the pattern was not confined to the foundation-model representation. Because demographic and technical differences are inseparable in this two-cohort design, the study does not assign the observed performance gap to a single cause.

2. Related Work

Recent evaluations have questioned whether single-cell foundation-model complexity consistently yields better downstream performance than simpler alternatives. Kedzierska et al. [10] found that zero-shot Geneformer and scGPT did not consistently outperform simpler baselines and remained sensitive to evaluation context. On a different task, Ahlmann-Eltze et al. [11] reported that deep-learning models did not outperform the linear baselines they evaluated for gene-perturbation prediction.

For scBERT, Boiarsky et al. [12] reported logistic regression performing comparably to or better than scBERT on their cell-type-annotation benchmarks and little effect of removing pretraining. Yang et al. [13] emphasized settings in which scBERT retained advantages, including cross-organ generalization and identification of novel or out-of-distribution cell types. These results show why conclusions about pretrained representations must remain task-specific.

The present study addresses the complementary question of transport after development choices are fixed. Clinical prediction methodology treats external validation as distinct from internal validation [5,6,14,15], while preregistration can constrain analytic flexibility and distinguish prespecified from post-hoc analyses [16,17]. Here those principles are applied to a donor-level single-cell foundation-model benchmark in SLE.

3. Methods

3.1 Cohorts

The development cohort was obtained from Perez et al. [7] through the CZ CELLxGENE Census [18] (collection 436154da-bcf1-4130-9c8b-120ff9a888f2, dataset 218acb0f-9f2f-4f76-b90b-15a4b7c7f629, CC BY 4.0; census version 2025-11-08). It comprised 261 adult donors (162 SLE, 99 healthy), aged 20–83 years, and 1,263,676 cells contributing to the corrected embeddings.

The external cohort was GSE135779 from Nehar-Belaid et al. [8]. Its public GEO series contains 56 samples: 33 SLE and 11 healthy in the source Children category, and 7 SLE and 5 healthy in Adult. Seven Children samples were aged 18 or 19, so source category names are retained. The source study describes 58 donors; aHD2 and aSLE8 are absent from the 56-sample public deposit, and the reason

remains unresolved. The 56 analyzed samples were 40 SLE and 16 healthy, aged 7–63 years. Cohorts differed in age structure and sequencing protocol. External sex and ancestry fields were not available in the analyzed GEO metadata; the study publication and supplementary workbook should be consulted for source-level clinical information [8].

Table 1. Cohort characteristics.

Characteristic	Development (Perez et al. [7])	External (Nehar-Belaid et al. [8], GSE135779)
Source / accession	CZ CELLxGENE Census, collection 436154da-...	NCBI GEO GSE135779
Donors, total	261	56 deposited; aHD2 and aSLE8 missing from the expected 58-label sequence
SLE / healthy	162 / 99	40 / 16
Age structure	All adult; 20–83 y (mean 41.3, SD 14.5; valid numeric age n = 259)	7–63 y (mean 20.9, SD 12.8); 44 source Children (including seven aged 18–19) / 12 Adult
Sex	244 female / 17 male	Not in public metadata
Self-reported ancestry	149 European American / 107 Asian / 3 African American / 2 Hispanic or Latin	Not in public metadata
Cells (approx.)	1,263,676 cells contributing to corrected embeddings	363,083 cells contributing to corrected embeddings
Gene reference	61,497 genes (Ensembl)	32,738 genes (Ensembl; 2019-era CellRanger v2 annotation per repository provenance)
Sequencing platform	Documented qualitatively in source publication	Different generation/protocol from development per source publications
Role	Development; internal nested cross-validation	Prespecified external validation

3.2 Feature arms

Geneformer V1 correction. The historical extraction selected V1 weights but omitted model_version=V1 and requested CLS embeddings, so those vectors are retained only as historical artifacts. The corrected extraction pinned Geneformer revision 04c2b2e84da7c0f385c3f9ad8f3ec24bab6650e5 and the V1-10M checkpoint, set model_version=V1, emb_mode=cell, emb_layer=-1, and model_input_size=2048, and averaged finite 256-dimensional cell embeddings by donor. The main input used the 30,165-gene reference intersection; unrestricted model-native input was exploratory. An eight-cell fixture checked dictionary loading, token IDs, special tokens, and output shape before extraction. The four archived run summaries record checkpoint and embedding hashes, cell accounting, and runtime packages. V1 dictionary hashes were printed during Kaggle setup but were not exported with those four share ZIPs; a later notebook revision exports machine-readable setup and checkpoint metadata.

Pseudobulk. Raw counts were summed within donor. Cross-cohort scoring used the 30,165 Ensembl genes shared by the 61,497-gene development reference and 32,738-gene external reference, representing 49.05% and 92.14% of those spaces. Counts were CPM-normalized using the restricted-gene total and log1p-transformed (deviations item 1). Both cohorts used Ensembl identifiers. The external reference is

recorded in the repository as a 2019-era CellRanger v2 annotation; this is repository provenance rather than a GEO-verified annotation version.

Metadata comparator. Donor age was the single feature; two development donors with ambiguous age records were excluded, leaving $n = 259$ (deviations item 5). All arms used StandardScaler followed by L2 LogisticRegression (lbfgs) in scikit-learn [21].

3.3 Donor-level nested cross-validation and inference

All analyses were donor-level, with one row per donor, preventing cells from the same individual from crossing folds and avoiding donor-level pseudoreplication [22,23]. Internal performance used five-fold nested cross-validation. Outer folds used StratifiedKFold(random_state=20 260 719); within each outer training fold, C was selected by five-fold grid search over [0.001, 0.01, 0.1, 1.0, 10.0, 100.0] without access to the outer test fold [24,25].

Pooled outer-fold donor predictions yielded internal AUROC. For the corrected Geneformer reanalysis, 5000 percentile bootstrap resamples of fixed donor predictions provided descriptive AUROC intervals. Baseline intervals and above-chance permutation p-values remain the historical released estimates where explicitly identified; they are not used to test equality of correlated AUROCs. Single-class bootstrap resamples were discarded.

Historical above-chance permutation procedures differed by arm: pseudobulk held $C = 0.01$ fixed while Geneformer and age repeated hyperparameter selection. These released p-values are preserved as historical results and do not provide paired comparisons. The corrected co-primary inference uses paired DeLong tests and paired donor-bootstrap intervals on aligned external donor predictions. With 1000 historical permutations, the smallest reported p-value was 1/1001 [26–29].

3.4 Preregistration and freeze

PREREG.md (tag prereg-locked-v1) specified the primary task, three arms, two co-primary comparisons, decision rule, Holm procedure, and confounders before development data were loaded. Before external access, FREEZE.json at commit 44d6f567 670 829 527 a6efa39c2d968c20e8d5515 recorded SHA-256 hashes for nine specified preregistration/development files and the prediction-determining development configuration and hyperparameters. It provides integrity checks for those files but does not establish that every later Phase 3 execution script was cryptographically frozen.

The historical frozen C was the median of five outer-fold selections: Geneformer 1.0 ([1, 1, 10, 10, 1]), pseudobulk 0.01 ([0.01, 0.1, 0.1, 0.01, 0.01]), and age 0.001 ([0.001]×5). Corrected shared-gene Geneformer development refitting selected [1, 10, 100, 10, 1] and median $C = 10$; native-input sensitivity selected [1, 10, 100, 100, 1] and median $C = 10$. The correction did not overwrite frozen or original release artifacts.

Two post-hoc development checks qualified the historical frozen hyperparameters. Across ten reseeded nested-CV runs on the original, unsupported Geneformer representation, the frozen C was recovered in 6/10 Geneformer, 0/10 pseudobulk, and 10/10 age runs. These are historical selection-stability results and do not assess the corrected V1 embeddings. Restricted-space pseudobulk nested CV selected [0.1, 0.1,

0.1, 0.01, 0.1] (median 0.1), whereas the frozen 0.01 had been selected in the full 61,497-gene space. The frozen 0.01 was retained for external pseudobulk scoring (deviations items 9–10).

3.5 External evaluation

For the corrected main Geneformer analysis, a scaler and L2 logistic regression were fitted on all 261 corrected development donor vectors using median nested-CV $C = 10$, then applied unchanged to the 56 corrected external vectors. Pseudobulk and age comparator probabilities were deterministically regenerated from the historical frozen analysis and checked for exact donor-key/label alignment. Development AUROC comes from pooled outer-fold predictions; external AUROC comes from a single full-development fit, so their difference is descriptive rather than a causal estimate of cohort shift.

The historical freeze guard was not called by every initial Phase 3 script: sealed pseudobulk processing and final scoring lacked `require_valid_freeze()` until later guard calls were added (deviations item 3). `SEALED_OPENED.json` records the initial external opening at 2026-07-21T06:50:41.257496+00:00. Historical external Geneformer scoring used frozen $C = 1$; the corrected V1 analysis reran development selection and used $C = 10$. Neither frozen historical artifacts nor the already-examined external sample status were changed by the correction.

3.6 Co-primary decision rule

The preregistered Geneformer-superiority rule required a paired donor-bootstrap 95% AUROC-difference interval wholly above zero and Holm-significant evidence for each comparison. The historical label-shuffle p-values tested above-chance label exchangeability, not equality of two correlated AUROCs. The correction therefore uses two-sided paired DeLong tests [37], applies Holm adjustment across the two comparisons, and retains 5000 paired donor-bootstrap intervals. DeLong inference is asymptotic and external controls number only 16. A positive difference and both statistical conditions are required for superiority; analyses on the previously examined external cohort are labeled corrected reanalysis.

3.7 Post-hoc analyses

The cohort-signature probe was pre-declared in PREREG.md §5.1 but deferred until after the co-primary results and is therefore treated as post-hoc (deviations items 2 and 7). New classifiers predicted cohort membership, not SLE status, from already-derived feature matrices and did not modify the frozen disease classifiers.

The versioned correction regenerated V1 development and external vectors using validated Kaggle outputs, repeated development nested CV, and rescored the already examined external cohort. The secondary analyses are age-stratified AUROC, average precision, Brier score, logistic recalibration, a development-prevalence constant-probability comparator, model-native input sensitivity, and an updated corrected-feature cohort-membership probe. These diagnostics do not create a new confirmatory cohort.

3.8 AI-assisted tools used during computational development

Anthropic Claude Sonnet 5 and OpenAI Codex assisted computational development, code review, and analysis documentation. The correction package was prepared for author verification; final author review and approval are required before journal submission.

4. Results

4.1 Internal and external performance

Table 2. Development and external performance.

Arm / feature space	Dev. n	Dev. AUROC (95% CI)	Historical above- chance p	External n	External AUROC (95% CI)	Historical above- chance p
Geneformer V1, shared genes	261	0.9746 (0.9548– 0.9901)	—	56	0.7344 (0.6017– 0.8597)	—
Pseudobulk, full developmen t space	261	0.9839 (0.9703– 0.9938)	0.000999	—	—	—
Pseudobulk, 30,165-gene space	261	0.9851 (0.9729– 0.9942), post-hoc	0.000999	56	0.8984 (0.8080– 0.9677)	0.000999
Age	259	0.6453 (0.5741– 0.7184)	0.000999	56	0.5781 (0.3614– 0.7815)	0.388

(Supplementary Table S1.)

The corrected Geneformer development-to-external point estimate changed by -0.2402 AUROC (0.9746 to 0.7344). The like-for-like restricted pseudobulk and age changes were -0.0867 and -0.0672 . These descriptive gaps compare nested-CV protocol estimates with full-development model applications and cannot isolate a cohort effect. Figure 1 shows the point estimates and their separate donor-bootstrap intervals.

Restricting development pseudobulk to the shared 30,165-gene space changed pooled development AUROC by $+0.00125$ (paired bootstrap 95% CI -0.00039 to $+0.00321$). Its fixed-C label-permutation $p = 0.000999$ tests above-chance discrimination in the restricted space; it does not test the restriction effect. Of 4015 genes absent across development donors, 490 were expressed in the external cohort; this descriptive availability check is not a causal explanation of the performance pattern.

4.2 External secondary analyses

In the source-defined Children group ($n = 44$; 33 SLE, 11 healthy), corrected shared-gene Geneformer AUROC was 0.7052 (95% CI 0.5452–0.8533); in the Adult group ($n = 12$; 7 SLE, 5 healthy), it was 0.8857 (0.6250–1.0000). The source Children category includes some donors aged 18 or 19. These exploratory stratum estimates are imprecise, especially for adults (Figure 2; Supplementary Table S2).

External average precision/Brier score was 0.899/0.471 for corrected shared-gene Geneformer, 0.962/0.149 for pseudobulk, and 0.744/0.216 for age. A constant probability equal to development SLE prevalence (162/261) had Brier score 0.213 externally. Diagnostic recalibration using the saved Geneformer decision_function scores gave intercept 2.091 (95% Wald CI 0.792–3.390) and slope 0.149

(0.027–0.271). Reconstructing logits after clipping eight saturated probabilities to $[10^{-6}, 1-10^{-6}]$ instead gives intercept 2.374 and slope 0.185. Pseudobulk and age slopes were 0.545 and -5.414 from clipped probability logits; one pseudobulk probability was clipped, and the age estimate was unstable. Calibration was diagnostic and did not change discrimination or evaluated probabilities.

4.3 Co-primary comparisons

Table 3. Corrected co-primary paired external comparisons (n = 56).

Comparison	AUROC difference	95% paired donor- bootstrap CI	DeLong p / Holm p	Rule met?
A: Geneformer – age	+0.1563	–0.0859 to +0.4040	0.2046 / 0.2046	Not met
B: Geneformer – pseudobulk	–0.1641	–0.2916 to –0.0384	0.0088 / 0.0175	Not met

The repository preserves the original preregistration, historical release, and released decision label. The revised manuscript uses “Not met” for a failed Geneformer-superiority criterion. Corrected result files are in results/correction_2026-09; the prior external cohort had already been examined.

Geneformer exceeded age by 0.1563 AUROC, but its paired CI included zero and Holm-adjusted p = 0.2046. Geneformer was 0.1641 AUROC below pseudobulk; the paired interval lay below zero and Holm-adjusted p = 0.0175. Neither contrast satisfies a positive Geneformer-superiority rule. The negative pseudobulk contrast is a result for this representation and cohort, not proof of broad model inferiority (Figure 3).

4.4 Cohort-signature probe

A post-hoc cohort-membership classifier using corrected shared-gene Geneformer vectors distinguished development from external donors with AUROC 0.9999 (95% CI 0.9996–1.0000). The preserved pseudobulk and age probes were 1.0000 (1.0000–1.0000) and 0.8719 (0.8021–0.9321), respectively. These are cohort-origin diagnostics, not SLE prediction or estimates of a causal batch contribution (Figure 4).

4.5 Prespecified cohort differences and confounders

Valid development age was available for 259 donors (mean 41.3, SD 14.5; range 20–83), all adults. External GEO samples had mean age 20.9 (SD 12.8; range 7–63), with 44 source Children and 12 source Adult records. Development metadata recorded 244 female/17 male donors and ancestry as 149 European American, 107 Asian, 3 African American, and 2 Hispanic or Latin. Corresponding fields were unavailable in the analyzed external GEO metadata. For aHD3 (GSM4029942, JB19003; 7,206 cells), GEO records age 50 while the study supplement was reported by an independent review to list 43; the workbook was unavailable for independent archival verification. We retained the preregistered GEO value for primary analysis and changed only this age to 43 in an exploratory fixed-model sensitivity. Overall age AUROC changed from 0.5781 to 0.5797, Adult age AUROC from 0.4286 to 0.4571, and Geneformer-minus-age from +0.1563 to +0.1547 (paired DeLong p = 0.2082; paired 95% CI -0.0862 to

+0.4022). The superiority conclusion was unchanged. Source studies document different sequencing protocols, but chemistry was not a harmonized structured field.

5. Discussion

5.1 Principal finding

Corrected Geneformer V1 yielded strong development discrimination (AUROC 0.9746) but did not meet either prespecified external superiority criterion. Its external AUROC was 0.7344 versus 0.8984 for pseudobulk and 0.5781 for age. The paired evidence supports a lower AUROC than pseudobulk for this specific corrected representation in this cohort; it does not establish general inferiority of Geneformer or clinical utility of any arm.

5.2 Cross-cohort performance pattern

The development-to-external AUROC point estimates differed by -0.240 for corrected Geneformer, -0.087 for restricted pseudobulk, and -0.067 for age. These comparisons are descriptive and combine changes in population, technical acquisition, and fitting procedure; they were not tested as causal cohort-shift effects.

The result is therefore narrower than a claim that Geneformer alone failed to transfer. Demographic composition, technical differences, cohort-associated structure, and the distinction between cross-validated protocol evaluation and final-model application remain intertwined.

5.3 Distribution shift

Cohort origin remained almost perfectly distinguishable using corrected Geneformer and pseudobulk features, while age alone was also informative for cohort membership. The diagnostic reveals strong cohort-associated structure but cannot identify its biological or technical source. The external cohort was mostly source-classified Children, while all development donors were adults; sex and ancestry metadata were unavailable externally. This limits attribution of the SLE-discrimination estimates.

Substantial demographic and technical distribution shift is consistent with known challenges in single-cell integration [31–33] and predictive-model transport [34,35]. Here the cohort-signature result supports treating the external population as meaningfully different while avoiding attribution of the performance pattern to one biological or modeling mechanism.

5.4 Implications for benchmarking

Three implications follow. First, claims of cross-cohort generalization should distinguish internal validation from evaluation in an independent population; a fixed analysis plan plus external data provides information that nested internal cross-validation cannot [5,6,14,15]. Sealing is one implementation of such precommitment, not a requirement in itself.

Second, development–external separability should be characterized alongside external performance. A cohort-signature diagnostic does not correct batch or demographic differences or establish causality, but it makes cross-dataset separability explicit and would be stronger if prespecified.

Third, model-selection stability should be reported separately from predictive stability. Pseudobulk AUROC was stable across reseeded nested CV, while the originally frozen $C = 0.01$ was recovered in 0/10 runs. This does not show that hyperparameter instability caused the external difference, but it exposes uncertainty hidden by a single cross-validation run. These considerations are not specific to foundation models; they are transparency checks for distinguishing within-cohort discrimination from performance in a different population.

5.5 Limitations

The external comparison has only 56 samples, including a 12-person source Adult subgroup. The public deposit lacks source-described aHD2 and aSLE8; why they are absent is unresolved. The GEO age 50 versus review-reported supplement age 43 for aHD3 is a source conflict rather than a parsing error, and the source workbook was unavailable for independent verification here. External outcomes had already been examined when corrected extraction and inference were performed. Thus, this is a corrected reanalysis, not untouched confirmation, and uncertainty intervals omit complete retraining and extraction uncertainty.

Cohort membership was strongly separable, but age structure, sequencing differences, unavailable external sex/ancestry, and possible pretraining overlap are entangled. The corrected -0.240 , -0.087 , and -0.067 development-to-external AUROC differences are descriptive and lack direct CIs for the gaps. This two-cohort design does not show that internal cross-validation is generally biased.

Preregistered cell-type harmonization and cell-type-resolved analyses were not performed because barcode-level labels for the exact analyzed raw-cell matrices and the planned label-transfer implementation were unavailable (deviations item 2). The source study's Supplementary Table 4 reports donor-level cluster and subcluster counts, which could support future composition analysis after sample and preprocessing reconciliation; those counts are not interchangeable with barcode-level labels. Prespecified age-stratified permutation p-values and stratified co-primary tests were not produced; revised age-stratified AUROCs are descriptive. Historical arm-level permutation procedures also differed in hyperparameter retuning.

Pseudobulk external scoring required the 30,165-gene intersection. The restricted-space development AUROC effect was small ($+0.00125$), but its above-chance permutation p-value is not an effect test. Model-native Geneformer input was an exploratory sensitivity analysis: external AUROC 0.7484 versus 0.7344 for shared-gene input (paired difference $+0.0141$, 95% CI -0.0238 to $+0.0597$; two-sided paired DeLong $p = 0.4601$). Neither native-input co-primary Geneformer-superiority criterion was met.

The original sealing mechanism supplied precommitment and file-integrity checks but did not technically block every historical external-path script. The correction preserved original artifacts and preregistered rules, recorded a new V1 checkpoint/runtime/feature provenance chain, and reanalyzed the already examined external data. Findings apply to frozen V1-10M cell embeddings, donor mean pooling, and L2 logistic regression. Other encoder versions, fine-tuning, cell-type strategies, cohorts, and possible Geneformer pretraining overlap were not assessed.

6. Conclusion

The corrected shared-gene V1 analysis yielded external AUROC 0.7344 and did not meet either prespecified Geneformer-superiority criterion under paired DeLong/Holm inference and paired donor-bootstrap intervals. The native-input sensitivity was similar. This corrected reanalysis does not constitute a new untouched validation on the previously examined external cohort.

The result does not establish that Geneformer is generally inferior, that pseudobulk is always preferable, or that any model is suitable for clinical diagnosis. It shows that one corrected frozen representation did not provide the prespecified added value in a small, distribution-shifted external cohort. Independent replication in another cohort is required for broader claims.

Declarations

Ethics approval and consent to participate. Not applicable. This study is a secondary computational analysis of previously published, de-identified, publicly available single-cell RNA-sequencing data (Perez et al. [7], via the CZ CELLxGENE Census; Nehar-Belaid et al. [8], NCBI GEO accession GSE135779). No new human-subject data were collected.

Consent for publication. Not applicable.

Data and code availability. Development data are available through the CZ CELLxGENE Census (collection 436154da-bcf1-4130-9c8b-120ff9a888f2; census version 2025-11-08), and external data through GEO GSE135779. Original preregistration and release commit 6158a840 remain preserved. Corrected extraction, validation, donor predictions, paired analyses, source metadata crosswalk, figures, and artifact hashes are versioned at <https://github.com/mohamad679/lupus-singlecell-foundation> on the scientific-correction-2026 branch. The corrected manuscript and supplement accompany the versioned submission package; public archival release remains subject to author approval. Supplementary Table S1 maps the numerical claims to artifacts.

Figure legends

Figure 1. Corrected shared-gene main analysis: development nested-CV and external fixed-model AUROCs for Geneformer V1, restricted pseudobulk, and age. Points are estimates and whiskers are separate 95% donor-bootstrap intervals; connecting lines are descriptive, not intervals for an internal-to-external effect. Geneformer development $n = 261$ and external $n = 56$; age development $n = 259$.

Figure 2. Corrected external donor-level AUROC by source-defined Children ($n = 44$) and Adult ($n = 12$) categories. Whiskers are 95% donor-bootstrap intervals. The Adult subgroup has 7 SLE and 5 healthy samples; some donors aged 18–19 are in the source Children category.

Figure 3. Corrected co-primary external comparisons. Points are paired Geneformer-minus-comparator AUROC differences and whiskers are 95% paired donor-bootstrap intervals. Displayed p-values are two-sided paired DeLong tests adjusted by Holm across the two comparisons. “Rule not met” denotes failure of the prespecified positive Geneformer-superiority conditions.

Figure 4. Post-hoc cohort-membership diagnostic using corrected shared-gene Geneformer vectors and preserved pseudobulk/age feature probes. The target is development versus external cohort origin, not SLE. The diagnostic quantifies feature-space separability without identifying a cause of disease-classification performance.

Declaration of generative AI and AI-assisted technologies in the manuscript preparation process

During preparation of the manuscript drafts, the author reported using Fable 5 for drafting and citation verification and OpenAI ChatGPT for drafting support, organization, language editing, consistency checking, and critical review. OpenAI Codex assisted this scientific correction and manuscript revision. The author should verify this disclosure, review the revised content, and accept responsibility before submission.

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